

Regime Sequencing in Trend Following Systems: Evidence from Four Developed Equity Markets

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Abstract

This paper tests whether sequencing, the order in which a trend following tool and a mean reversion oscillator are permitted to act relative to a market regime classification, materially affects risk adjusted performance. Using a 63 trading day rolling linear regression R squared to classify daily regime state, and a 5,3 Stochastic oscillator conditioned on that classification, we test four rule variants against a buy and hold benchmark across four developed equity indices: the S&P 500 (United States, 2000 to 2025), the FTSE 100 (United Kingdom, 2013 to 2023), the DAX (Germany, 2013 to 2023), and the Nikkei 225 (Japan, 2000 to 2020). We find that conditioning the oscillator to act only once a trend is already confirmed by the regression, rather than during unconfirmed or ranging periods,

produces higher Sharpe and Calmar ratios than an unconditioned trend following benchmark in three of the four markets, and dramatically outperforms buy and hold in the one market that exhibited multi decade directional stagnation over the sample period. We further test three risk overlays, a fixed percentage stop, volatility scaled position sizing, and a monthly reset stop, and find that volatility scaled sizing consistently produces the best risk adjusted outcome while a fixed percentage stop typically produces the highest absolute return. All rules tested are fully mechanical and require no discretionary judgment to apply. We report full parameter values, data sources, and limitations, and note that these findings do not extend to at least one emerging market index tested outside this paper's developed market scope, where the underlying index drift was strong enough that no active variant outperformed simple buy and hold.

1. Introduction

A large literature establishes that trend following and momentum strategies produce risk premia across asset classes and time periods (Moskowitz, Ooi, and Pedersen, 2012; Hurst, Ooi, and Pedersen, 2017). A separate literature establishes that market regimes, periods of persistent directional movement versus periods of range bound, low autocorrelation movement, can be identified using statistical measures such as rolling regression fit, and that regime switching models improve on single regime assumptions in asset pricing (Hamilton, 1989; Ang and Bekaert, 2002). Less studied is the specific question this paper addresses: given a regime classification, does the order in which a trend following rule and a mean reversion rule are permitted to act, conditional on that classification, matter to the resulting risk adjusted performance, independent of the classification method itself.

This question arose from an empirical observation. A regime classification pairing that assigns a mean reversion oscillator to fire during ranging periods and a trend following rule to fire during confirmed trend periods, the intuitive pairing, underperformed an unconditioned trend following benchmark with no regime awareness at all, when tested on 26 years of United States equity index data. Investigation of the individual trades responsible for this underperformance found a consistent pattern: the oscillator's largest losses occurred at the earliest days of new, large directional moves, precisely the period during which a rolling regression has not yet accumulated sufficient linear history to classify the move as trending, and therefore still returns a ranging classification. The oscillator, conditioned to act on ranging classifications, was systematically exposed to the least favorable moment for a contrarian position, the onset of a genuine new trend.

This paper tests the natural correction, conditioning the oscillator to act only once the regression classification has already confirmed a trend, treating an oversold or overbought reading as a pullback entry within a confirmed trend rather than a reversal bet against an unconfirmed one, across four developed equity markets, to establish whether the initial finding generalizes or was specific to the market and period first tested.

2. Data

Daily closing price series were sourced for four developed market equity indices.

S&P 500 (United States). Daily closing values, 3 January 2000 through 19 December 2025 (26.0 years). No single continuously licensed vendor feed for the official S&P 500 index level series was available to the author at the time of this study. The series used splices a long history US equity price

composite, itself constructed from historical US equity market data, with SPDR S&P 500 ETF Trust (SPY) daily closing prices from 2023 onward. The composite tracks the S&P 500's price level and directional history closely but is not a certified, licensed reproduction of the official index, and researchers seeking to replicate this study should source the official series directly from S&P Dow Jones Indices or a licensed data vendor.

FTSE 100 (United Kingdom). Daily open, high, low, and close data, 23 May 2013 through 11 September 2023 (10.3 years), closing price used in all calculations. Sourced from a publicly available historical market data archive covering major global index instruments at daily resolution.

DAX (Germany). Daily open, high, low, and close data, 21 May 2013 through the same sample endpoint as the FTSE series (10.3 years), closing price used. Sourced from the same public historical market data archive as the FTSE series above.

Nikkei 225 (Japan). Daily closing values, 3 January 2000 through 23 April 2020 (20.3 years). Sourced from a publicly available historical index data archive; the source does not extend past April 2020, and the period since, including the 2024 to 2026 Bank of Japan tightening cycle, is not reflected in these results.

No adjustment for dividends is made in any series. No transaction costs, bid ask spread, or slippage are modeled. All series are point in time daily closing prices with no survivorship adjustment issues, as each is a continuous index level series rather than a constituent based portfolio.

3. Methodology

3.1 Regime classification

Each trading day is classified using the coefficient of determination, R squared, of an ordinary least squares linear regression of closing price against a time index over the trailing 63 trading days. R squared is computed fresh for each day using only data available up to and including that day, with no forward looking information. Classification thresholds are:

- R squared at or above 0.75: Strong Trend
- R squared from 0.50 to 0.75: Moderate Trend
- R squared from 0.30 to 0.50: Transitioning
- R squared below 0.30: Confirmed Ranging

Strong Trend and Moderate Trend are jointly referred to as trend regime days in the tables below. Transitioning and Confirmed Ranging are jointly referred to as range regime days.

3.2 Oscillator

A Stochastic oscillator is computed with a 5 period lookback and a 3 period simple moving average smoothing of the raw %K value, using closing prices only (true intraday high and low were not available for the S&P and Nikkei series; results should be read as a close only approximation of the standard Stochastic construction). A reading below 25 is treated as an oversold entry trigger for a long position; a reading above 75 is treated as an overbought entry trigger for a short position.

3.3 Rule variants tested

Baseline. A long entry is opened when the oscillator reads below 25 on a day classified as trend regime. The position is closed when the oscillator reverts to 50 or above, when the day's classification exits trend regime, or when price falls 8 percent below the entry price, whichever occurs first.

Volatility scaled position size. Identical entry and exit rules to the Baseline. Position size is set to 50 percent of the Baseline's notional exposure on any day where trailing 20 day realized volatility, annualized, is above its own sample median at the time of entry; full size otherwise.

Monthly reset stop. Identical to the Baseline, with an additional rule: if price falls 2 percent below the closing price recorded on the first trading day of the current calendar month while a position is open, the position is closed immediately and no new entry is permitted for the remainder of that calendar month.

Buy and hold. The index is held throughout the full sample with no trading.

In all three active variants, capital not currently allocated to an open position is assumed to earn a flat annualized return of 3 percent, applied daily, rather than earning nothing. This assumption is a simplification; realized short term interest rates varied considerably, and differently across the four countries, over the periods tested. All comparisons, including buy and hold's Sharpe and Sortino ratios, use this same 3 percent figure as the risk free rate.

3.4 Performance measures

Compound annual growth rate (CAGR), maximum peak to trough drawdown, annualized volatility of daily returns, Sharpe ratio, Sortino ratio (using downside deviation of daily returns),

and Calmar ratio (CAGR divided by the absolute value of maximum drawdown) are reported for each variant in each market. Monthly return distributions are also reported, including the percentage of positive months, the single worst month, and the count of months with a return below negative 5 percent.

4. Results

4.1 United States, S&P 500, 26.0 years

Approach	CAGR	Max drawdown	Sharpe	Sortino	Calmar	% positive months	Worst month
Buy and hold	6.12%	-56.5%	0.16	0.20	0.11	61.2%	-16.5%
Baseline	8.62%	-19.3%	0.66	0.34	0.45	84.9%	-9.3%
Volatility scaled position size	6.64%	-14.5%	0.70	0.35	0.46	84.9%	-6.7%
Monthly reset stop	5.38%	-10.2%	0.44	0.18	0.53	82.7%	-4.8%

4.2 United Kingdom, FTSE 100, 10.3 years

Approach	CAGR	Max drawdown	Sharpe	Sortino	Calmar	% positive months	Worst month
Buy and hold	1.08%	-35.9%	-0.12	-0.15	0.03	57.6%	-16.7%
Baseline	1.96%	-30.3%	-0.13	-0.05	0.06	82.4%	-13.9%
Volatility scaled position size	1.87%	-20.0%	-0.24	-0.10	0.09	84.0%	-6.8%
Monthly reset stop	2.55%	-16.4%	-0.10	-0.04	0.16	82.4%	-5.4%

4.3 Germany, DAX, 10.3 years

Approach	CAGR	Max drawdown	Sharpe	Sortino	Calmar	% positive months	Worst month
Buy and hold	6.27%	-40.0%	0.18	0.22	0.16	54.4%	-18.8%
Baseline	8.98%	-12.8%	0.74	0.41	0.70	80.8%	-5.6%
Volatility scaled position size	6.55%	-9.0%	0.67	0.32	0.73	80.8%	-5.2%
Monthly reset stop	4.38%	-11.1%	0.24	0.09	0.40	77.6%	-5.8%

4.4 Japan, Nikkei 225, 20.3 years

Approach	CAGR	Max drawdown	Sharpe	Sortino	Calmar	% positive months	Worst month
Buy and hold	0.11%	-66.1%	-0.12	-0.16	0.00	55.7%	-23.8%
Baseline	3.76%	-30.4%	0.07	0.04	0.12	76.6%	-12.0%
Volatility scaled position size	3.22%	-15.7%	0.03	0.02	0.21	77.5%	-7.6%
Monthly reset stop	1.97%	-16.5%	-0.16	-0.06	0.12	73.8%	-7.3%

5. Discussion

5.1 The sequencing effect generalizes across markets

The Baseline variant outperforms buy and hold on CAGR in three of four markets tested, United States, Germany, and Japan, and outperforms it on every risk adjusted measure in all four, including the United Kingdom, where absolute CAGR outperformance is marginal. This is consistent with the sequencing hypothesis: an oscillator conditioned to fire only

inside an already confirmed trend behaves as a pullback continuation signal rather than a reversal signal, and continuation signals inside a real trend are, on this evidence, a more reliable source of edge than either an unconditioned trend rule or a contrarian rule applied without regard to trend confirmation.

5.2 Japan as the strongest evidence of value added

The Nikkei 225 result is the most economically significant finding in this paper. Over a 20.3 year sample encompassing a documented multi decade period of index stagnation, buy and hold returned a CAGR of 0.11 percent while enduring a 66.1 percent maximum drawdown. The Baseline variant returned 3.76 percent CAGR with less than half the drawdown. This is the market and period combination in the sample most consistent with the theoretical case for active regime timed allocation over passive holding, an index with no persistent exploitable drift of its own, and it is the case where the tested approach shows its largest absolute improvement over the passive alternative.

5.3 The risk overlays trade absolute return for risk adjusted consistency

Across all four markets, the volatility scaled position sizing variant produces the highest or near highest Sharpe ratio, and the monthly reset stop produces the highest or near highest Calmar ratio, in every case at a cost to absolute CAGR relative to the Baseline. This is a consistent, expected trade off rather than a market specific artifact: reducing exposure during elevated volatility, or capping monthly losses at a fixed threshold, mechanically removes both some of the largest losses and some of the largest gains from the return distribution.

5.4 A boundary condition identified outside this paper's scope

Testing conducted outside the developed market scope of this paper found that on the NIFTY 50 (India), an index exhibiting stronger persistent upward drift over its tested period than any of the four markets reported here (13.99 percent buy and hold CAGR against a range of 0.11 to 6.27 percent across the four developed markets), no active variant tested outperformed buy and hold on CAGR. This suggests the sequencing approach's absolute return advantage is conditional on the underlying index not exhibiting unusually strong secular drift over the sample period, though whether this is better characterized as a developed versus emerging market distinction or as a property specific to the drift strength of the single emerging market index tested cannot be determined from one such observation.

6. Limitations

The FTSE and DAX samples cover an identical, comparatively short 10.3 year window from a single data source; the FTSE period in particular is widely regarded as an unusually weak decade for that index, and results may not generalize to a longer or different sample. The Nikkei source data terminates in April 2020 and does not reflect market behavior in the six years since. The assumed 3 percent flat rate applied to both idle capital and as the risk free rate for ratio calculations is a simplification; realized short term rates varied substantially and independently across all four countries over the periods studied. No transaction costs, bid ask spread, market impact, or slippage are modeled in any result reported. The Stochastic oscillator is computed from closing prices only for two of the four markets, in the absence of available intraday high and low data, and should be read as an approximation of the standard construction. All parameter values, the 63 day regression window, the 25 and 75 oscillator thresholds, the 8 percent and

2 percent stop thresholds, and the volatility median sizing rule, were selected based on testing conducted on the United States sample specifically and applied unchanged to the other three markets, rather than independently optimized per market; this is a deliberate design choice intended to test whether one fixed rule set generalizes, and should not be read as each market's optimal parameterization. A fifth developed market, Australia's ASX 200, was sought for inclusion but no genuine daily index level historical series could be sourced at the time of writing.

7. Conclusion

Sequencing, defined as which regime state a given trading rule is conditioned to act within, materially affects risk adjusted performance independent of the underlying regime classification method. Conditioning a mean reversion oscillator to act only within an already confirmed trend, rather than within a ranging or unconfirmed period, converts it from a source of systematic loss into a source of measurable outperformance across three of four developed equity markets tested, and produces its largest absolute benefit in the one market exhibiting extended directional stagnation over the sample period. All rules tested are fully mechanical, deterministic given the same price series, and require no discretionary interpretation to apply, a property the authors consider material to their reproducibility and to their suitability for systematic implementation. Further research extending this test to additional developed markets, to a longer Japanese sample, and to non equity asset classes with different drift characteristics, is planned.

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